Department of Computer Engineering

National Institute of Technology Kurukshetra

Mid Term- 2, Even Semester, 2024-2025

Subject: Deep Learning (CSTC 308/ITTC 308) **Branch:** B.Tech. *CS and IT (VI Semester)*
Maximum Marks: 15 **Time:** 50 Minutes **Date:** 08/04/2025 **Roll No:** _____

Note: Kindly don't make stories. Avoid unnecessary elaboration, as meaningless content may negatively impact marking.

1. Why are scaling (γ) and shifting (β) often applied after the standard normalization in the batch normalization layer? **[2 marks]**
2. Prove that adding noise to the inputs acts as a regularization technique in neural networks. Provide a mathematical explanation to support your argument. **[3 marks]**
3. Consider a convolutional neural network (CNN) designed for digit recognition of numbers 0-9, using a dataset of color images with a resolution of 28×28. The network begins with a convolutional layer that applies 32 filters of size 3×3, with a stride of 1 and no padding. This is followed by a pooling layer with a 2×2 filter, a stride of 1, and no padding. Next, the network includes a fully connected layer with 64 neurons, followed by the output layer, which is also a fully connected layer responsible for classifying the *digits from 0 to 9*. Determine the total number of parameters in the network and specify the output size after each layer. **[4 marks]**
4. I have been asked to train a neural network for a binary classification task. I used a sigmoid non-linear activation function at the output node to model the probability of classification. Instead of using cross entropy error, I used sum of squared error for computing the loss for back propagation. Do you foresee any problem with this choice of loss function? Briefly explain. [Hint: Check the gradient for both the loss functions with respect to the output layer weights] **[6 marks]**